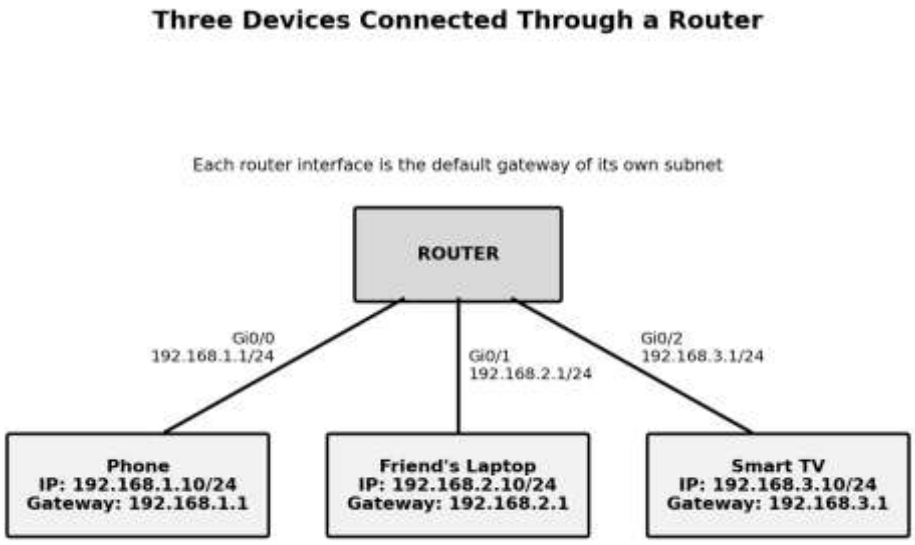
IP Routing and Routing Protocols

Assignment Answers

Q1. Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.



Device / Interface	IP address	Subnet	Role
Phone	192.168.1.10/24	192.168.1.0/24	Connected to router interface Gi0/0 (default gateway 192.168.1.1)
Friend's Laptop	192.168.2.10/24	192.168.2.0/24	Connected to router interface Gi0/1 (default gateway 192.168.2.1)
Smart TV	192.168.3.10/24	192.168.3.0/24	Connected to router interface Gi0/2 (default gateway 192.168.3.1)
Router Gi0/0	192.168.1.1/24	192.168.1.0/24	Gateway for the phone's subnet
Router Gi0/1	192.168.2.1/24	192.168.2.0/24	Gateway for the laptop's subnet
Router Gi0/2	192.168.3.1/24	192.168.3.0/24	Gateway for the smart TV's subnet

Q2. Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made. Use made-up IP addresses and show the routing table entries that would be checked at each hop.

Setup (made-up addresses)

- My laptop: 192.168.1.10/24, default gateway 192.168.1.1 (Router R1, interface Gi0/0).
- Router R1: Gi0/0 = 192.168.1.1/24 (my LAN), Gi0/1 = 10.0.0.1/30 (link to R2).
- Router R2: Gi0/1 = 10.0.0.2/30 (link to R1), Gi0/0 = 192.168.2.1/24 (friend's LAN).

- Friend's computer: 192.168.2.20/24, default gateway 192.168.2.1.

Path: Laptop (192.168.1.10) to R1 to R2 to Friend's computer (192.168.2.20).

Step 1: At my laptop

The laptop compares the destination with its own network: 192.168.2.20 AND 255.255.255.0 = 192.168.2.0, which is not equal to its own network 192.168.1.0. So the destination is in a different subnet. The laptop therefore sends the packet to its default gateway 192.168.1.1. It finds the router's MAC address using ARP and builds a frame with destination MAC = R1 and destination IP = 192.168.2.20.

Step 2: At Router R1 (first hop)

R1 receives the frame, removes the Layer 2 header and reads the destination IP 192.168.2.20. It checks its routing table and selects the entry with the longest matching prefix:

Destination network	Mask	Next hop	Interface	Type
192.168.1.0	255.255.255.0	Directly connected	Gi0/0	C (connected)
10.0.0.0	255.255.255.252	Directly connected	Gi0/1	C (connected)
192.168.2.0	255.255.255.0	10.0.0.2	Gi0/1	S (static)

The destination 192.168.2.20 matches the entry 192.168.2.0/24, so the next hop is 10.0.0.2 (R2) through Gi0/1. R1 reduces the TTL by 1 (for example 64 to 63), builds a new frame with R2's MAC address and forwards the packet.

Step 3: At Router R2 (second hop)

Destination network	Mask	Next hop	Interface	Type
192.168.2.0	255.255.255.0	Directly connected	Gi0/0	C (connected)
10.0.0.0	255.255.255.252	Directly connected	Gi0/1	C (connected)
192.168.1.0	255.255.255.0	10.0.0.1	Gi0/1	S (static)

R2 checks the destination 192.168.2.20 and finds that it matches the directly connected network 192.168.2.0/24 on Gi0/0. It reduces the TTL again (63 to 62), finds the friend's MAC using ARP and delivers the frame to the friend's computer.

Step 4: At the friend's computer and the reply

The friend's computer receives the packet and accepts it, because the destination IP is its own. The reply goes back by the same process in the reverse direction: friend's computer to R2 (its gateway) to R1 to my laptop, using the entries 192.168.1.0/24 via 10.0.0.1 on R2 and 192.168.1.0/24 connected on R1.

Key points

- The source and destination IP addresses stay the same during the whole journey, but the MAC addresses change at every hop.
- The routing decision is made by the longest prefix match in the routing table; if no entry matches, the packet goes to the default route or is dropped.
- The TTL decreases by 1 at every router, which prevents packets from looping forever.

Q3. Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

Fictional network: A head office (HQ) router connects three city branches, and each branch has its own subnet. All addresses are made-up examples.

Branch / subnet	Network	Link to HQ	Router interface	Purpose
HQ (head office)	10.1.0.0/24	-	Gi0/0	HQ servers and admin
Mumbai branch	10.1.1.0/24	172.16.1.0/30	Gi0/1	Mumbai restaurants and delivery partners
Delhi branch	10.1.2.0/24	172.16.2.0/30	Gi0/2	Delhi restaurants and delivery partners
Bengaluru branch	10.1.3.0/24	172.16.3.0/30	Gi0/3	Bengaluru restaurants and delivery partners

Routing table of the HQ router

Network address	Subnet mask	Next hop	Interface	Route type
10.1.0.0	255.255.255.0	Directly connected	Gi0/0	Connected
172.16.1.0	255.255.255.252	Directly connected	Gi0/1	Connected
172.16.2.0	255.255.255.252	Directly connected	Gi0/2	Connected
172.16.3.0	255.255.255.252	Directly connected	Gi0/3	Connected
10.1.1.0 (Mumbai)	255.255.255.0	172.16.1.2	Gi0/1	Static
10.1.2.0 (Delhi)	255.255.255.0	172.16.2.2	Gi0/2	Static
10.1.3.0 (Bengaluru)	255.255.255.0	172.16.3.2	Gi0/3	Static
0.0.0.0 (default)	0.0.0.0	203.0.113.1 (ISP)	Gi0/4	Static default route

Routing table of the Mumbai branch router (example)

Network address	Subnet mask	Next hop	Interface	Route type
10.1.1.0	255.255.255.0	Directly connected	Gi0/0	Connected
172.16.1.0	255.255.255.252	Directly connected	Gi0/1	Connected
0.0.0.0 (default)	0.0.0.0	172.16.1.1 (HQ)	Gi0/1	Static default route

Explanation: The HQ router sends traffic for each city through the correct link to that branch. The branch routers use a default route towards HQ, so the traffic for all other cities and the internet goes through the head office.

Q4. Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

Static routing (routes entered manually)

Pros:

- Low overhead: it does not use extra CPU, memory or bandwidth because no routing updates are exchanged.
- Predictable and more secure: the path never changes unless the admin changes it. Example: a home or a small shop's router uses one default route to the ISP.

- Simple to set up in a small network, such as a single branch office with one connection.

Cons:

- Manual work: every change in the network must be updated by hand on every router.
- No automatic recovery: if a link fails, traffic does not move to another path until the admin fixes it. A food delivery app would have orders stuck during an outage.
- Does not scale: it becomes difficult and error-prone when there are many routers and networks.

Dynamic routing (RIP, OSPF)

Pros:

- Automatic updates: routers share information and learn new networks without manual work, which suits a growing network with many branches.
- Automatic failover: if a link fails, the routers find another path. For example, a food delivery app with several data centres can keep taking orders if one link goes down.
- Scalable: it works well for large networks. OSPF chooses the best path using cost (bandwidth) and is used in college campuses and ISPs such as those serving Jio or Airtel users.

Cons:

- Higher overhead: routers use CPU, memory and bandwidth for exchanging updates.
- More complex to configure and troubleshoot, and it needs a skilled administrator.
- Less secure if not protected: a wrong or fake routing update can send traffic to the wrong place. RIP is also slow to react and limited to 15 hops.

RIP vs OSPF (short comparison)

Point	**RIP**	**OSPF**
Type	Distance vector	Link state
Metric	Hop count (maximum 15)	Cost based on bandwidth
Speed of convergence	Slow	Fast
Best for	Small and simple networks	Medium to large networks

Q5. Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does. Paste the AI-generated config and annotate each part with a short description.

The sample configuration below was generated using an AI tool (Claude) and then explained line by line. It is written for Cisco IOS routers. Topology: R1 and R2 are connected by the link 10.0.0.0/30. R1 has the LAN 192.168.1.0/24 and R2 has the LAN 192.168.2.0/24.

Configuration of Router R1

Configuration line	**What it does**
`Enable`	Enters privileged mode of the router.
`configure terminal`	Enters global configuration mode, where settings can be changed.
`hostname R1`	Gives the name R1 to the router so it is easy to identify.
`interface GigabitEthernet0/0`	Selects the interface connected to the LAN.
`ip address 192.168.1.1 255.255.255.0`	Gives the LAN interface its IP address and subnet mask; this is the gateway for the LAN devices.
`no shutdown`	Turns the interface on (interfaces are off by default).

Configuration line	What it does
`interface GigabitEthernet0/1`	Selects the interface connected to Router R2.
`ip address 10.0.0.1 255.255.255.252`	Gives the link interface its IP address; /30 allows only 2 usable hosts, which is enough for a link between two routers.
`no shutdown`	Turns this interface on.
`router ospf 1`	Starts the OSPF process with process ID 1. The ID is only local to this router and need not match the other router.
`router-id 1.1.1.1`	Sets a unique identity for this router in OSPF.
`network 192.168.1.0 0.0.0.255 area 0`	Tells OSPF to run on interfaces in 192.168.1.0/24 and advertise this network to neighbours. 0.0.0.255 is the wildcard mask (the opposite of 255.255.255.0). Area 0 is the backbone area.
`network 10.0.0.0 0.0.0.3 area 0`	Enables OSPF on the link to R2 (wildcard 0.0.0.3 matches /30), so R1 can form a neighbour relationship with R2.
`passive-interface GigabitEthernet0/0`	Stops OSPF messages on the LAN interface (no routers there), which is safer and saves bandwidth, but the LAN is still advertised.
`End`	Returns to privileged mode.
`copy running-config startup-config`	Saves the configuration so that it is not lost after a restart.

Configuration of Router R2

R2 uses the same commands with these different values:

Configuration line	What it does
`hostname R2`	Name of the second router.
`interface GigabitEthernet0/0`	Selects the LAN interface of R2.
`ip address 192.168.2.1 255.255.255.0`	LAN interface of R2 with its own network 192.168.2.0/24.
`interface GigabitEthernet0/1`	Selects the link interface towards R1.
`ip address 10.0.0.2 255.255.255.252`	Link to R1; it is in the same /30 network as R1's interface 10.0.0.1.
`router ospf 1`	Starts the OSPF process (same process ID is fine, but it is only local).
`router-id 2.2.2.2`	Gives R2 a different, unique router ID.
`network 192.168.2.0 0.0.0.255 area 0`	Advertises R2's LAN in area 0.
`network 10.0.0.0 0.0.0.3 area 0`	Enables OSPF on the link, so R1 and R2 become neighbours in the same area.
`passive-interface GigabitEthernet0/0`	No OSPF messages on the LAN side.

Checking that it works

- show ip ospf neighbor: shows whether R1 and R2 have become neighbours (state FULL).
- show ip route ospf: shows the routes learned by OSPF (marked with the letter O); R1 should show 192.168.2.0/24 and R2 should show 192.168.1.0/24.
- ping from a PC in 192.168.1.0/24 to a PC in 192.168.2.0/24 to confirm end-to-end connectivity.

Result: Both routers automatically exchange their routes using OSPF, so no static routes need to be written by hand.